

AYMANE AIT DADS

Machine Vision Intern | Computer Vision · Intelligent Systems · Data Pipeline Engineering

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PROFESSIONAL SUMMARY

Developed and deployed an end-to-end machine vision classification pipeline at Orange Maroc processing over 100,000 fiber-optic network records, achieving a 60% reduction in manual analysis time through automated feature engineering and model evaluation. Ranked 1st in EURECOM class on a computer vision benchmark (0.791 AUC) using CLIP ViT-L/14 fine-tuned on 250K images across 25+ generator types, and submitted to the NTIRE 2026 @ CVPR international benchmark. Core competencies include computer vision, image classification, sensor data processing, intelligent system design, Python, PyTorch, and end-to-end ML pipeline deployment. Available immediately for a European internship; based in Sophia Antipolis, France, with full authorization to work in Europe.

CORE COMPETENCIES

Machine Vision | Data Collection | Intelligent Systems | Sensor Deployment | Image Classification | Software Tooling | Hands-on Hardware-Data Integration | End-to-End Pipeline

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built an end-to-end machine vision and data pipeline combining Random Forest and K-Means on **100,000+ fiber-optic network records** to classify network quality across thousands of sensor nodes.
- Developed an automated clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team.
- Reduced stakeholder manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and structured reporting in Power BI.
- Delivered technical presentations translating model outputs into **actionable maintenance recommendations** for non-technical stakeholders, bridging intelligent system outputs and business decisions.

PROJECTS

AI-Generated Image Detection - EURECOM ImSecu + NTIRE 2026 @ CVPR

EURECOM Course + International Benchmark · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class leaderboard (0.791 AUC)** on a binary image classification task using CLIP ViT-L/14 (428M params) fine-tuned on 250,000 images spanning 25+ generator types; also submitted independently to NTIRE 2026 @ CVPR CodaBench international benchmark.
- Engineered a **3-model ensemble with 10-view TTA** (flips, crops, blur, resizes) and dual learning rates (2e-6 backbone, 5e-4 head), discovering that 1-epoch training outperforms 4-epoch by 2.6 AUC points for out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle T4 GPU

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition (In Progress)

Kaggle Competition · June 2026 · \$106K+ Prize Pool · Top 10% Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** on a \$106K+ prize-pool competition fine-tuning a 30B Mamba-Transformer MoE model using LoRA (r=32) with ~888M trainable parameters on 7,828 verified chain-of-thought examples for 6 logic puzzle types.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict dependency order - including monkey-patched CUDA allocators, stubbed Mamba imports, and pure PyTorch fallbacks - and conducted 20+ single-variable ablations confirming completion-only loss and packing configuration as decisive factors.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · Triton · vLLM

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, NLP, Statistics, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Computer Vision & ML Frameworks: PyTorch | OpenCLIP | CLIP ViT | CNN | DenseNet | Scikit-learn | TensorFlow

Data Pipeline & MLOps: Python | Pandas | NumPy | Feature Engineering | Git | FP16 Mixed Precision | Power BI

LLM & NLP Tooling: Hugging Face Transformers | LoRA / PEFT | SFTTrainer | vLLM | Sentiment Analysis | SQL